

EXP 6

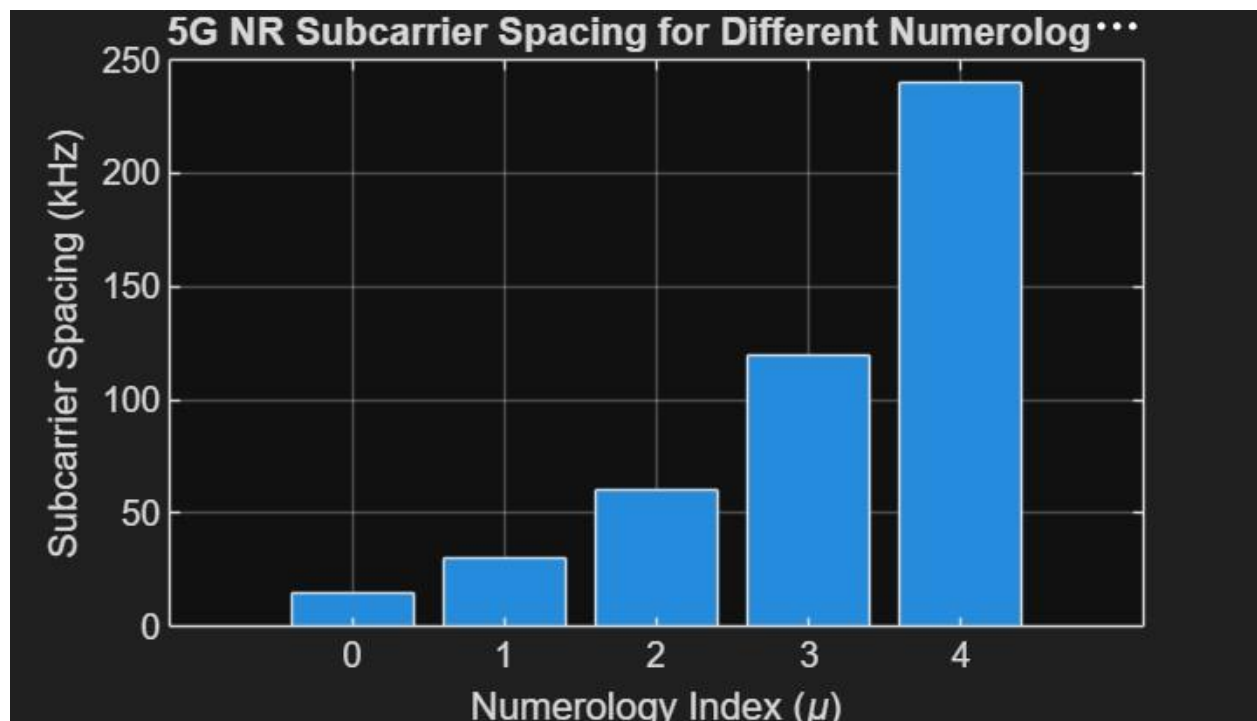
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OBJECTIVE 1:

CODE:

```
clc;
clear;
close all
;
scs = [15 30 60 120 240]; % Subcarrier Spacing
(kHz) mu = 0:4; % Numerology Index bar(scs) grid
on set(gca, 'XTick',1:5) set(gca, 'XTickLabel',mu)
xlabel( 'Numerology Index (\mu)' ) ylabel(
'Subcarrier Spacing (kHz)' )
title( '5G NR Subcarrier Spacing for Different Numerologies' )
```

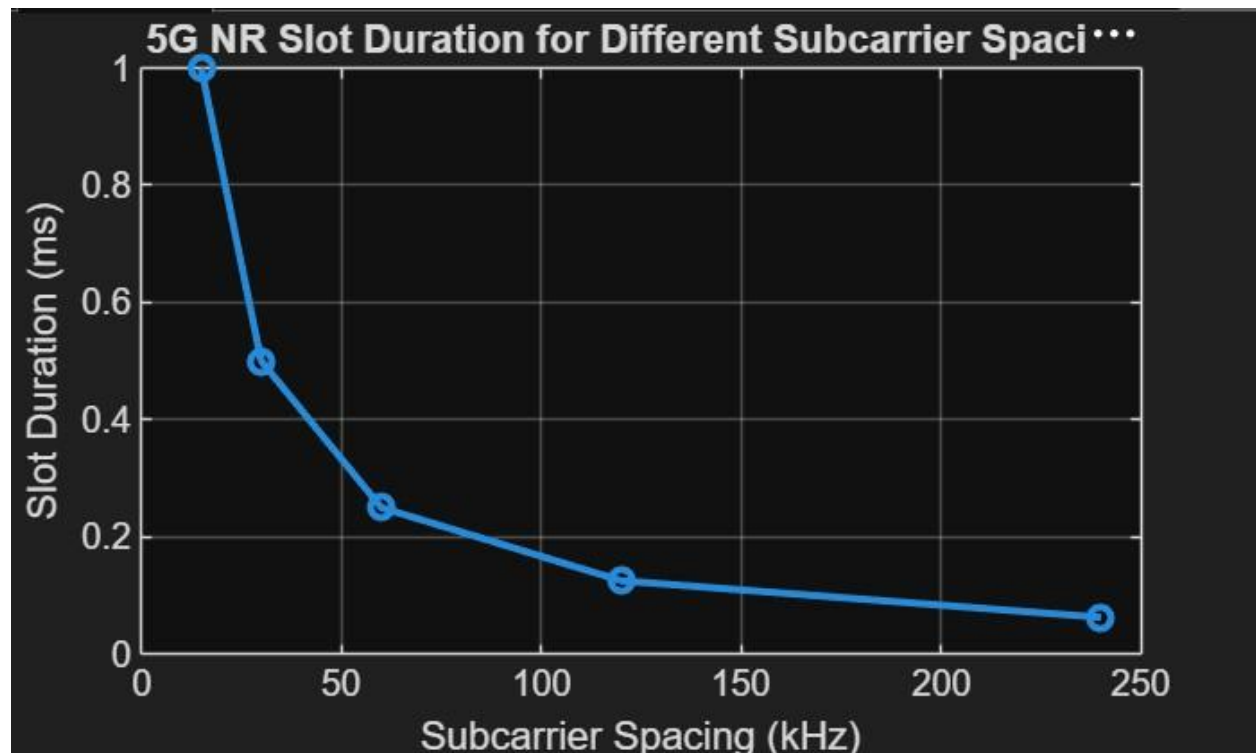


OBJECTIVE 2:

CODE:

```
clear;  
close all  
;  
scs = [15 30 60 120 240]; % Subcarrier Spacing  
(kHz) slot = [1 0.5 0.25 0.125 0.0625]; % Slot  
Duration (ms) plot(scs, slot, 'o-', 'LineWidth', 2)  
grid on xlabel('Subcarrier Spacing (kHz)') ylabel(  
'Slot Duration (ms)')  
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

OUTPUT:



OBJECTIVE 3:

CODE:

```

clc;
clear;
close all
;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure bar(scs, symbols) grid on xlabel('Subcarrier Spacing (kHz)') ylabel('Number of OFDM Symbols') title('OFDM Symbols per Slot for Different Subcarrier Spacing') ylim([0 16])

```

OUTPUT:

